Atomistic Modelling of Molecules and Materials (MLE5215) Fall 2025

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Force Fields & Geometry Optimisation Schemes

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DISCLAIMER: These notes are far from perfect and your input during this class is highly appreciated.

1 Conservation of Energy, Kinetic and Potential Energies

We shall now examine one of the most basic laws of physics, *the conservation of energy*, which states:

In an *isolated system*, i.e., a system that does not exchange energy and matter with its surroundings, the *total energy of the system is conserved.* The total quantity of energy of an isolated system does not change, though it may change form.

The principle of the *conservation of energy* is central to our understanding and the rest of this course. Concerning newtonian or classical physics, in general, the conservation laws include the conservation of energy, mass, linear momentum, angular momentum, and electric charge. Some forms of energy include:

Very important for the rest of this class are the concepts of *Kinetic energy* T and *Potential energy* U, which we revise briefly here.

Kinetic energy— The kinetic energy, T, of an object of mass m, moving with a velocity $\mathbf{v}$ is given in Eq. 1.1.

$$T = \frac{1}{2}m|\mathbf{v}|^2 \tag{1.1}$$

Potential energy— Its definition is a little less straightforward compared to the kinetic energy of Eq. 1.1. Since it is an energy of location (or position) concerning some reference point, the potential energy of an object depends on the specific situation. An example is gravity. An object of mass m at a height h above the Earth's surface experiences a gravitational potential energy V defined in Eq. 1.2.

$$V = mgh \tag{1.2}$$

where g is the acceleration due to the Earth's gravity. In this example, the potential energy V has a simple form, namely that it depends linearly on the height h of the object. V may even be varying in time if, for example, the object is falling toward the Earth. In general, the potential energy may not be such a simple function of location. In this case, one needs a potential energy curve to describe the potential energy as a function of some coordinate describing the object's location at any point concerning the earth's surface.

A more pertinent example of potential energy for this class is the definition of potential energy $V_{i,j}(r)$ between atoms i and j at a distance r in a diatomic molecule, as shown in Figure 1.

The chemical forces that hold the molecule together come from a potential energy curve $V(r)$ that depends only on the distance between the hydrogen atoms. The presence of a deep "well" indicates a particular distance of lowest potential energy —this is also called a *global minimum* in the potential energy. This corresponds to the bond length of the molecule and is, therefore, the most likely value of the separation of the two atoms.

If a force F acting on an object is a function of position only, it is said to be a *conservative force*, and it can be represented by a potential energy function V which for a one-dimensional case satisfies the derivative condition of Eq. 1.3

$$F(x) = -\frac{dV}{dx} \tag{1.3}$$

Thus, the force on an object *is the negative* of the derivative of the potential function V. This means it is the negative of the slope of the potential energy curve.

If Eq. 1.3 is integrated the corresponding potential is given by Eq. 1.4.

$$V(x) = -\int_{x_0}^{x} F(x)dx + V(x_0) \tag{1.4}$$

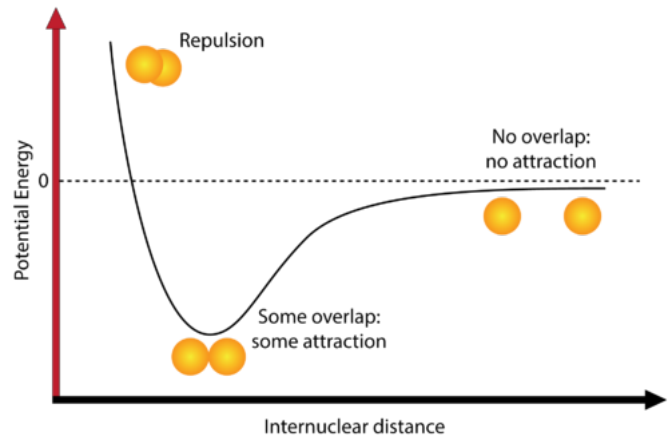

Figure 1: The potential energy $V_{i,j}(r)$ of two hydrogen atoms in H_2 changes as a function of the separation between atoms, i.e., the distance r.

which can be taken as a definition of potential energy. Note that there is an arbitrary constant of integration in that definition, showing that any constant can be added to the potential energy. Practically, this means that you can set the zero of potential energy at any convenient point.

Therefore, the potential energy V is equal to the work one must perform to move an object from the reference $V = 0$, i.e., x_0 point to the position x. The reference point at which one assigns the value $V = 0$ is arbitrary, so it may be chosen conveniently to simplify the problem.

A further observation can be made on the sign of the potential energy V. Conventionally, the potential energy V is equal to the work one must exert against that force to move an object from the $V = 0$ reference point to the position r. The force one must exert to move the object needs to be equal, but oppositely directed. This explains the source of the negative sign in the definition of the potential V. Thus, the force exerted by the force field always tends toward lower energy and will act to reduce the potential energy.

In light of these definitions and observations, we conclude with a general statement: the total energy E of a system (i.e., an atom, a molecule, or a solid) *must be conserved* (it is set to a value) through a balance between the kinetic and potential energy terms, as displayed in Eq. 1.5.

$$E = K + V \tag{1.5}$$

Eq. 1.5 is central to the entire class.

1.1 The Chemical Bond: The Morse and Buckingham Potentials

Figure 2 shows the simulated potential of H_2 through advanced quantum chemical methods, which are the object of the following lectures. The black vertical line in Figure 2 shows the bond length of H_2 measured experimentally ~ 0.743 Å.

The computed minima (black circles in Fig. 2) is ~ 0.737 Å, which is remarkably close to the experimental value (with a relative error from the experiment of only $\sim 0.8\%$).

However, these results can be fitted by embarrassingly simple analytical functions "incorporating" the main physics of the chemical bond and yet very accurate. These functional forms are the *Morse* and *Buckingham* potentials that include both the repulsive terms at very small intermediate distances, as well as the attractive van der Waals forces at long interatomic distances (see Fig. 1).

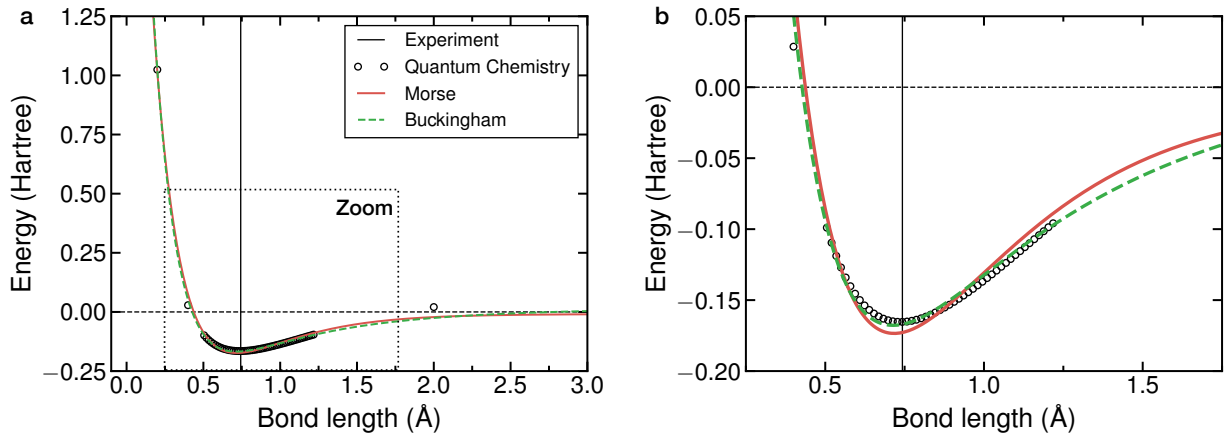

Figure 2: Panels **a** and **b** simulated potential of H_2 (black circles) and fitted with a number of physical models, i.e. Morse (red solid line) and Buckingham (green dashed line) potentials, respectively. Panel **b** is a zoom of the bond zone in H_2 around the experimentally determined value.

In the following sections we will replace the notation V introduced in Eq 1.4 to define the potential into E. This is because we assume that the positions of the nuclei in molecules or materials are at equilibrium and hence are not subjected to kinetic energy, but solely potential energy.

The Morse potential of Eq. 1.6 has the peculiar feature that describes accurately that when the bond is stretched to very large distances, the potential energy $E(r)$ should approach the value of dissociation energy of the molecule in its respective atoms.[1]

$$\begin{aligned} E_{\text{Morse}}(x - x_0) &= D\left[1 - e^{-\alpha(x-x_0)}\right]^2 \\ \alpha &= \sqrt{\frac{k}{2D}} \end{aligned} \tag{1.6}$$

where D is the dissociation energy of the bond, α is the force constant (the spring constant) of the bond and x_0 is the equilibrium bond length. The Morse function reproduces quite accurately the computed potential of Fig. 2 (red solid line) over a wide range of distances. However, there are a number of difficulties with the Morse potential in real applications. For example, for long bond lengths, the restoring force is small, and thus highly distorted structures will display slow convergence toward the equilibrium bond length.

The Buckingham potential of Eq. 1.7 and Fig. 2 (dash green line) is another effective analytical function to describe the behavior of bonds in molecules as shown by its remarkable agreement with the high-level quantum chemistry data.

$$E_{\text{Buckingham}}(x - x_0) = Ae^{-\frac{x-x_0}{\rho}} - \frac{C}{(x - x_0)^6} \tag{1.7}$$

where A, ρ and C are parameters that are varied to reproduce the experimental data. Differently from the Morse potential (Eq. 1.6), the Buckingham potential contains an explicit definition of the attractive and repulsive parts. Noticeably, the right-hand side term $\frac{1}{x^6}$ describes the attractive Van der Waals interaction at long bond lengths, see Fig. 2 **a**. The left-hand side of Eq. 1.7 describes the repulsive part at short bond lengths and it is called the Born-Mayer potential. The Buckingham potential form is employed in many force fields

[1] The dissociation energy of a $\text{H}-\text{H}$ bond is simply defined as $\Delta E_{\text{diss}} = 2E_{\text{H}} - E_{\text{H}_2}$, where E_{H} and E_{H_2} are the total energies of the atomic and molecular hydrogen, respectively.

and interatomic potentials (see Sec. 3). When C in Eq. 1.7 is set to 0 we have the Born-Mayer type potential.

Here, we learned that the chemical bond between two atoms can be treated explicitly by simple functional forms, such as the Morse and the Buckingham potentials. Furthermore, since these potentials are additive, one can imagine computing the total energy of a system (i.e., molecule, protein, or a solid) by *breaking it down into many single diatomic* Morse/Buckingham/Born-Mayer potential interaction. We will return to this concept.

2 The Potential Energy Surface

Earlier, Figures 1 and 2 helped us to re-introduce the concept of potential energy V. These figures are also commonly known as the *potential energy surfaces (PESs)*. The PES for a diatomic homopolar molecule, as in the case of H_2 (or O_2, N_2, etc.) is embarrassingly simple, and zones, where Coulomb repulsion and Van Der Waals attraction dominate, are easily identified.

However, when one deals with molecules or solids of interest with sizes ranging from tens to millions of atoms, the PES can become extremely complicated (see Figure 3) to the point that its graphical representation is cumbersome (as it would require many dimensions), i.e., $3\times number\ of\ atoms$.

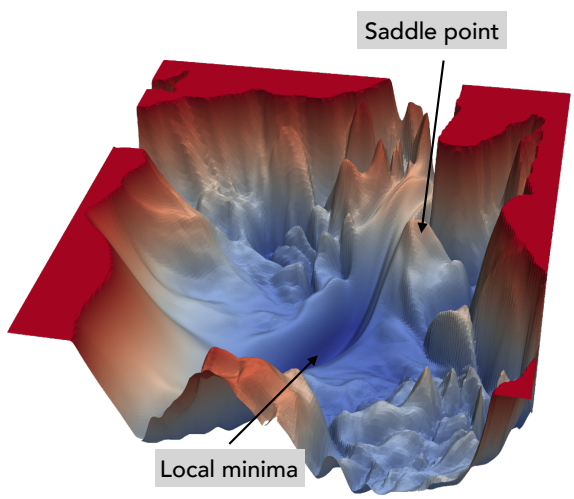

Figure 3: Complex potential energy surface PES of a molecule including a multitude of local minima (blue) and saddle points (red).

Importantly, the PES can have several minima in energy, which correspond to local equilibrium positions (specific location of atoms in space). Local minima abound in chemical systems and materials displaying a lot of degrees of freedom; for example, biological structures, including enzymes, polymers, etc. Positive ridges identify saddle points in PESs, connect local minima, and corresponds mathematically to saddle points. Saddle points in the PES represent *transition states* —which are of importance in studying chemical reactions.

The optimization techniques to identify local and global minima in PES will be discussed in Sec. 5.

Strictly speaking, the dimensionality of a PES —the number of dimensions defining the PES— depends entirely on the number of atoms in the system. The dimensionality of the PES is given by:

$$\mathrm{PES}_{\mathrm{dimension}} = 3 \times N - 6 \tag{2.1}$$

where N is the number of atoms, whereas the factor 3×N makes sure that we are considering the x, y, z coordinates of each atom. Finally, -3 is to remove the 3 trivial translations of the system, and another -3 for trivial rotations. Now imagine that we are interested in a simple molecule, such as benzene with a chemical formula of C_6H_6, hence 12 atoms. The dimensionality of the PES associated with benzene will be 30.

3 Force Fields & Interatomic Potentials

Given a certain molecule or a material, the challenge is to describe its PES accurately so that one can map its stable configurations and possible transition states. At a given nuclear configuration (meaning at a given arrangement of atoms in space) the challenge is to calculate the electronic energy of the system to map the PES.

In *force filed* (FF) methods, the electronic energy is written as a parametric function of the positions of the nuclei in the molecule or materials. The parameters entering the parametric/empirical functions are fitted from existing experimental data or high-level computational data.

When one deals with materials rather than molecules, the term force field is replaced by *interatomic potential*. For the sake of simplicity, we will use the term force field interchangeably.

Importantly, the *building blocks* in FF are atoms and the effect of the electron is not considered explicitly. This signifies that information on boding between atoms must be provided *explicitly*, rather than being the results of some more complex models (e.g., the Schrödinger equation). In FF molecules are described by a *ball and spring* model, with atoms having different sizes and *softness* and bonds having different lengths and *stifness*.

Therefore the total energy or internal energy (E_{Total}) of the system[2] is approximated by the force field total energy E_{FF} of Eq. 3.1, which is written as a sum of terms, each describing the energy required to distort a molecule specially.

$$E_{\text{Total}} \approx E_{\text{FF}} = E_{\text{stretching}} + E_{\text{bending}} + E_{\text{torsion}} + E_{\text{vdW}} + E_{\text{electrostatic}} + E_{\text{cross}} \tag{3.1}$$

where $E_{\text{stretching}}$ is the energy required to stretch the bond between two atomic species, the E_{bending} the energy to bend an angle involving three atoms, E_{torsion} is the energy required to rotate a bond of a solid angle. The E_{vdW} and $E_{\text{electrostatic}}$ terms describe the non-bonded atom-to-atom interactions. Finally, E_{cross} accounts for the cross interaction of $E_{\text{stretching}}$, E_{bending}, and E_{torsion}, respectively.

Some of the energy terms of Eq. 3.1 can be visualized in Fig. 4.

To define the E_{FF} of Eq. 3.1 one needs to have some knowledge of the connectivity between atoms in the system under investigation.

3.1 The Stretching Energy Term

$E_{\text{stretching}}$ is the energy required to stretch the bond between two atoms M and N. To define $E_{\text{stretching}}$ one can resort to the Taylor expansion around the equilibrium distance x_0 (e.g., the minimum in Fig. 2 for H_2). If one truncates the Taylor expansion at the second order obtains:

$$E_{\text{stretching}}(x^{\text{MN}} - x_0^{\text{MN}}) = E(0) + \frac{dE}{dx}(x^{\text{MN}} - x_0^{\text{MN}}) + 0.5\frac{d^2E}{dx^2}(x^{\text{MN}} - x_0^{\text{MN}})^2 \tag{3.2}$$

[2]Since the nuclei are at rest, the kinetic energy $K = 0$, and hence the total energy is equal to the potential energy $E_{\text{FF}} = V$.

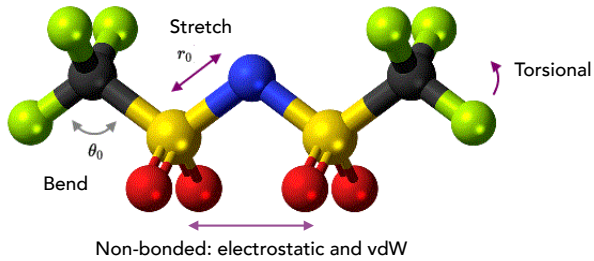

Figure 4: Graphical representation of some of the energy terms taking part to the formulation of the FF, as detailed in Eq. 3.1.

where the derivatives are typically evaluated at $x = x_0^{\mathrm{NM}}$ and $E(0)$ set to 0. The terms $\frac{dE}{dx}(x^{\mathrm{MN}} - x_0^{\mathrm{MN}})$ is zero as the 1st derivative at a minimum is always zero, and thus the $E_{\mathrm{stretching}}$ becomes:

$$E_{\mathrm{stretching}}(x^{\mathrm{MN}} - x_0^{\mathrm{MN}}) = k_2^{\mathrm{MN}}(x^{\mathrm{MN}} - x_0^{\mathrm{MN}})^2 = k_2^{\mathrm{MN}}(\Delta x^{\mathrm{MN}})^2 \tag{3.3}$$

where $k_2^{\mathrm{MN}} = \frac{d^2E}{dx^2}$ in Eq. 3.3 is the force constant associated to the M—N bond. The shape of the PES described in Eq. 3.3 is harmonic or quadratic and reflects the harmonic response of the bond upon stretching (see harmonic approximation in lecture slides). The harmonic treatment of the bond is the simplest possible and sufficient in determining most equilibrium geometries.

Nevertheless, the harmonic potential is not always sufficient and a natural improvement requires the inclusion of higher order terms in the Taylor expansion as in Eq. 3.4.

$$E_{\mathrm{stretching}}(x^{\mathrm{MN}} - x_0^{\mathrm{MN}}) = k_2^{MN}(\Delta x^{\mathrm{MN}})^2 + k_3^{\mathrm{MN}}(\Delta x^{\mathrm{MN}})^3 + k_4^{\mathrm{MN}}(\Delta x^{\mathrm{MN}})^4 + \cdots \tag{3.4}$$

The price to pay with an increase in accuracy is the introduction of additional parameters, which do not always have physical meaning. In addition, polynomial expansion does not follow the correct asymptotic behavior. For example, the cubic k_3 is typically negative, and if the Taylor expansion is truncated at the third order, the Energy will approach $-\infty$ at increasing bond lengths, which is not expected if one looks at the typical potential curve of a bond, see, for example, Fig. 2.

In Sec. 1.1, and precisely Figure 2**a**, we have introduced the Morse potential, which in contrast to Eq. 3.4 behaves appropriately at long bond-distances reproducing correctly the dissociation limit. Furthermore, the functional form of the morse potential is significantly simpler than Eq. 3.4 and hence advantageous for its implementation in computer simulation codes.

3.2 The Bending Energy Term

Assuming that there are bonds between L-M and M-N, the energy required to bend an angle θ between 3 atoms L-M-N is E_{bending}. Again, we resort to the Taylor expansion of E_{bending} and truncate it to the second order (see Eq. 3.5)

$$E_{\mathrm{bending}}(\theta^{\mathrm{LMN}} - \theta_0^{\mathrm{LMN}}) = k^{\mathrm{LMN}}(\theta^{\mathrm{LMN}} - \theta_0^{\mathrm{LMN}})^2 \tag{3.5}$$

In analogy to what is said for $E_{\mathrm{stretching}}$ in Sec. 3.1 the accurate description of E_{bending} may require higher terms than the second order. Similar issues as discussed in the case of the polynomial expansion of $E_{\mathrm{stretching}}$ are to be expected for higher-order terms in the Taylor expanded E_{bending}.

3.3 The Torsion Energy Term

Assuming that there exist a solid angle L-M-N-O and three bonds (i.e., L-M, M-N, and N-O), the energy required to impart a torsion of a dihedral angle ϕ between 4 atoms L-M-N-O is E_{torsion}. A visual representation of a dihedral angle ϕ in the organic molecule ethane is shown in Fig. 5.

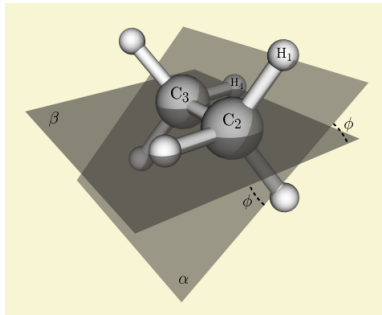

Figure 5: Graphical representation of the dihedral angle ϕ in the organic molecule ethane formed between two imaginary planes α and β and interesting some atoms of interest.

Noticeably, the torsion energy is fundamentally different from $E_{\text{stretching}}$ and E_{bending} at least in three aspects:

1. A rotational barrier of a solid angle is affected by both the torsional energy itself and non-bonding (i.e., vdW and electrostatic, see Eq. 3.1) contributions, as the relative position of the atoms involved in the torsion changes with respect to the other atoms (not bonded) in the surrounding. Hence, the torsional energy is intimately coupled to the non-bonding interactions.

2. The function of the potential energy associated with E_{torsion} must be periodic in nature. In other words, subsequent 360° rotations of the angle ϕ in Figure 5 will map the same potential energy landscape.

3. The energy penalty paid to distort a molecule around a bond is often low or *soft*. Thus, Taylor expansions of E_{torsion} are not recommended.

The smartest way to cast the periodic profile of E_{torsion} is in terms of a Fourier series, as in Eq. 3.6.

$$E_{\text{torsion}}(\phi) = \sum_{n=1} V_n \cos(n\phi) \tag{3.6}$$

where n is the period of the rotation, and $n = 1$ for 360°, $n = 2$ for 180°, $n = 3$ for 120°, etc. The constant V_n sets the size of the barrier to rotate the molecule around the bond of the solid angle, e.g., the bond C_2-C_3 in Figure 5.

In many practical cases the value of $V_n = 0$; for example, it is experimentally known that in the case of ethane (Figure 5), the most stable conformation is the one where the hydrogen atoms on C_2 and C_3 are staggered relative to each other, while the eclipsed conformations (where H_1 and H_4 of Figure 5 face each other) form a maximum in the PES of ethane. As the three hydrogen atoms on each carbon atom are undistinguishable, there will be three identical staggered minima (where $V_n = 0$ kJ mol^{-1}) and three identical eclipsed maxima (where $V_n \sim 12$ kJ mol^{-1}, as depicted in Figure 6).

In the Fourier series only those therms $n = 3, 6, 9$, etc., will have $V_n \neq 0$.

3.4 Non-Bonding Terms: Van der Waals & Electrostatic

E_{vdW} and $E_{\text{electrostatic}}$ in Eq. 3.1 deal with the Van der Waals and electrostatic attractive and repulsive interaction, hence these terms are not describing interactions rising from chemically bonded atoms. The E_{vdW} may be interpreted as the description of all the interplay between atoms arising from non-polar interactions, which are *not* accounted for by the Coulombic term $E_{\text{electrostatic}}$. An example is the interaction of two ethene molecules that not including the polar group will be only attracted by London's or Van der Waals forces.

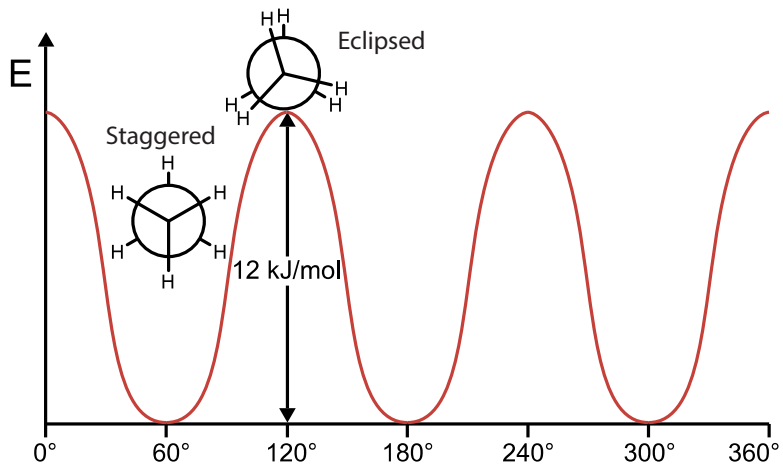

Figure 6: Periodic potential energy surface of $E_{\text{torsion}}(\phi)$ in the organic molecule ethane, showing the three eclipsed and staggered configurations, respectively.

The Van der Waals term —E_{vdW} becomes zero at very large distances (e.g., 10 Å) and becomes very repulsive at short-distances. These interactions are due to instantaneous dipole-dipole correlations stemming from electron correlation phenomena and hence fluctuate over time. Even if the molecule does not have a permanent dipole moment, as, in the case of ethane, the oscillatory nature of electrons will produce uneven distributions of the electronic charge. Such instantaneous polarisation, will induce an instantaneous polarisation in the nearby molecules causing attraction.

The induced dipole-dipole interactions are only the leading term of such induced multipole interactions: there are also higher-order contributions from induced dipole-quadrupole, quadrupole–quadrupole, dipole-hexadecapole interactions, etc. However, such interactions vary as r^{-8}, r^{-10}, etc., and the r^{-6} dependence is only the asymptotic behaviour at long distances. The force associated with this potential is often referred to as a "dispersion" or London's force.

The E_{vdW} assumes very positive values (unstable) at small distances and displays a minimum that is slightly negative at a distance corresponding to the two atoms just "touching" each other, whereas E_{vdW} becomes asymptotically null at a very large distance. A general functional form that fits these conditions is given by Eq. 3.7.

$$E_{\text{vdW}}(x_{\text{MN}}) = E_{\text{repulsion}}(x_{\text{MN}}) - \frac{C_{\text{MN}}}{x_{\text{MN}}^6} \tag{3.7}$$

Eq. 3.7 appears very similar to the Buckingham potential in Eq. 1.7. It is impossible to derive theoretically the functional form of the repulsive interaction, but it is only required that it goes toward zero as x (the interatomic distance) goes to infinity, and it should approach zero faster than the x^{-6} term, as the energy should go towards zero from below.

However, a function that obeys these general requirements is the Lennard-Jones (LJ) potential, where the repulsive part is given by an x^{-12} dependence and C_1 and C_2 are fitted constants.

$$E_{\text{LJ}}(x_{\text{MN}}) = \frac{C_1}{x_{\text{MN}}^{12}} - \frac{C_2}{x_{\text{MN}}^6} \tag{3.8}$$

From electronic structure theory it is known that the repulsion is due to the overlap of the electronic wave functions, and that the electron density falls off approximately exponentially with the distance from the nucleus —the exact wave function for the hydrogen atom is an exponential function. This statement provides enough justification to treat the repulsive part using an exponential function, and the x^{-6} exponential-like function is exactly cast in the Buckingham model we introduced in Eq. 1.7, where A, ρ and C are constant parameters which are varied to reproduce experimental data.

Finally, it is worth noting that the Morse potential has also an exponential functional form, hence the correct shape as seen in Figure 2.

The Electrostatic term —The other part of the non-bonded interaction is due to the internal (re)distribution of the electrons, creating positively and negatively charged parts of the molecule. An OH^- group, for example, has a negatively charged oxygen and a positively charged hydrogen. At the lowest approximation, this can be modeled by assigning (partial) charges to each atom that is +1 to hydrogen and -2 to oxygen. Alternatively, one can assign a dipole moment (and higher-order terms) to the bond. These two descriptions give similar (but not identical) results. Only in the long-distance limit of interaction between such atoms/-molecules do the two descriptions give identical results.

For brevity, here we only describe the Coulomb term of Eq. 3.9. The interaction between point charges M and N is given by the Coulomb potential, with ε the dielectric constant

$$E_{\text{Coulomb}}(x_{\text{MN}}) = \sum_{M,N}^{n} \frac{Q_M Q_N}{\varepsilon x_{\text{MN}}} \tag{3.9}$$

where n is the number of atoms. The atomic charges can be assigned by empirical rules. As long one respects the charge neutrality condition —the simulation box as a total charge of zero— it is plausible to assign fractional charges to species. This is often the case of inter-atomic potentials used to describe materials. Another example where partial charges are useful is when describing hydrogen bonding, which is to a large extent due to attraction between the electron-deficient hydrogen and an electronegative atom such as oxygen or nitrogen. Thus, a proper choice of partial charges may adequately model this interaction.

In general, large values ε screen (buffer) the effect of the naked charges. Thus, a proper choice of dielectric constant ε in Eq. 3.9 can be included to model the effect of surrounding molecules (e.g., solvent) and interactions between distant sites, which may be "via" parts of the same molecule. This latter strategy is useful to introduce polarisation effects in a mean-field, and often the case of large biological macromolecules, such as enzymes. A value of $\varepsilon = 1$ corresponds to a vacuum, while larger values of ε reduce the importance of long-range charge-charge interactions. Typically, a value between 1 and 4 is used, although there is little theoretical justification for any specific value. In some applications, the dielectric constant is made distance dependent (e.g., $\varepsilon = \varepsilon_0 x_{\text{MN}}$, and hence changing the Coulomb interaction to $\frac{Q_M Q_N}{\varepsilon x_{\text{MN}}^2}$) to model "screening" effects imparted by solvent molecules.

In the case of solid materials where an infinite number of atomic species is considered, Eq. 3.9 becomes crucial. In the case of a solid Eq. 3.9 run to infinity and hence divergent series in real space. The technique proposed by Peter Ewald —the so-called Ewald-summation— invoking the reciprocal space allows an exact solution for this problem.

Final remark We have expressed the mathematical form of many important interactions to capture meaningful physics happening in molecular and materials systems. These terms contribute to the definition of E_{FF} given in Eq. 3.1 which is the force field or interatomic potential used to approximate the total energy of the system.

3.5 Popular Force Fields and Their Transferability

We have just discovered a way to describe molecules and materials bypassing the Shrödinger equation, whereby the electrostatic energy is developed in terms of parametric functions of the sole nuclear coordinates. Such functions are called force fields (or inter-atomic potential in materials science). Once a finite number of equations capturing the appropriate level of physics is captured, the force field parameters are fitted to experimental or higher-level computational data.

Historically, in 1977 Norman L. Allinger proposed one of the first force fields [J. Am. Chem. Soc., 99, 25, 8127-8134 (1977) doi:10.1021/ja00467a001], called *molecular mechanics* MM, whose successive versions are MM2 and MM3.

Transferability of FF —Again, the fundamental idea behind force-field methods lies in the observation that molecules tend to be composed of units that are structurally similar in different molecules. For example, all the C—H bond lengths are roughly constant in all molecules, being between 1.06 and 1.10 Å. The frequency associated with the C—H stretching vibration is also similar, ranging between 2900 and 3330 cm^{-1}, implying that the associated force constants are also comparable. If the C—H bonds are further divided into groups, for instance, those attached to a single-, double- or triple-bonded carbon, the variation within each of these groups become even smaller. The same grouping strategy holds for other structural types, e.g., all C=O bonds are approximately 1.22 Å long, and vibrate around 1700 cm^{-1}, all double-boded carbon is virtually planar, etc. The *transferability* also holds for energetic data.
Types in force fields —The idea of molecules being composed of atoms, which are structurally similar in different molecules is implemented in force fields as atom *types*. Strictly speaking, the atom type depends on the atomic number and the type of chemical bonding formed. The type may be denoted with either a number or a letter code. For example, in MM2 there are 71 different atom types (note, type 44 is missing). Type 1 is an sp^3-hybridised carbon, whereas a hybridised sp^2 carbon may be type 2, 3 or 50, depending on the neighbor atom(s). Type 2 is used if the carbon atom is bonded to another sp^2 carbon (through a simple double bond C=C). Type 3 is when a carbon atom is double bonded to an oxygen atom as in C=O, while type 50 is used if the carbon atom is part of an aromatic ring with a delocalized bond, such as in benzene. Figure 7 gives a complete list of the MM2(91) atom types.
Different Force Fields —Since MM2, several force fields have been proposed in the literature and they differ in two main aspects:

- What is the functional form of each energy term?
- What types of information are used for fitting the parameters?

There are two general trends.
Class I —If the force field is designed primarily to treat large molecular systems, such as proteins or DNA,[3] the functional forms are kept as simple as possible. In which case only harmonic functions are used for $E_{\text{stretching}}$ and E_{bending} (or these terms are omitted, forcing all bond lengths and angles to be constant), and the Lennard-Jones potential is used for E_{vdW}. Such force fields are often called "harmonic", "diagonal" or *Class I*.
Class II —The second approach deals with reproducing small- to medium-size molecules to a high degree of accuracy. These force fields will include several cross terms, use at least cubic or quartic expansions of $E_{\text{stretching}}$ and E_{bending}, and possibly an exponential-type potential for E_{vdW}. The current efforts in developing *Class II* force fields go in the direction of not only striving to reproduce geometries and relative configurational energies, but also vibrational frequencies.
United Atoms —Force fields designed to treat macromolecules can be simplified by excluding hydrogens explicitly in the so-called united atom approach —an option present in some popular FF, such as AMBER, CHARMM, GROMOS, and DREIDING. In these force fields, instead of modeling a CH_2 group as a carbon and two hydrogens, a single CH_2 atom may be assigned, and such a *united atom* will have a larger Van der Waals radius to account also for the hydrogens. The advantage of implementing *united atoms* is that they effectively reduce the number of variables by a factor of ∼2-3, thereby allowing correspondingly larger systems to be treated as the overall computational cost is decreased. Of course the coarser the atomic description is, the less detailed the final results will be. Which description, and thus which type of force field to use depends entirely on what type of information is sought. If the interest is in geometries and relative energies of different conformations of a small molecule, such as hexane, then an elaborate force field (e.g.,

[3] Note, the nuclear position of large molecular systems of biological interest are often determined through standard techniques of X-ray diffractions or nuclear magnetic resonance.

Type	Symbol	Description	Type	Symbol	Description
1	C	sp^3-carbon	28	H	enol or amide
2	C	sp^2-carbon, alkene	48	H	ammonium
3	C	sp^2-carbon, carbonyl, imine	36	D	deuterium
4	C	sp-carbon	20	lp	lone pair
22	C	cyclopropane	15	S	sulfide (R_2S)
29	C·	radical	16	S^+	sulfonium (R_3S^+)
30	C^+	carbocation	17	S	sulfoxide (R_2SO)
38	C	sp^2-carbon, cyclopropene	18	S	sulfone (R_2SO_2)
50	C	sp^2-carbon, aromatic	42	S	sp^2-sulfur, thiophene
56	C	sp^3-carbon, cyclobutane	11	F	fluoride
57	C	sp^2-carbon, cyclobutene	12	Cl	chloride
58	C	carbonyl, cyclobutanone	13	Br	bromide
67	C	carbonyl, cyclopropanone	14	I	iodide
68	C	carbonyl, ketene	26	B	boron, trigonal
71	C	ketonium carbon	27	B	boron, tetrahedral
8	N	sp^3-nitrogen	19	Si	silane
9	N	sp^2-nitrogen, amide	25	P	phosphine
10	N	sp-nitrogen	60	P	phosphor, pentavalent
37	N	azo or pyridine (–N═)	51	He	helium
39	N^+	sp^3-nitrogen, ammonium	52	Ne	neon
40	N	sp^2-nitrogen, pyrrole	53	Ar	argon
43	N	azoxy (–N═N–O)	54	Kr	krypton
45	N	azide, central atom	55	Xe	xenon
46	N	nitro ($–NO_2$)	31	Ge	germanium
72	N	imine, oxime (═N–)	32	Sn	tin
6	O	sp^3-oxygen	33	Pb	lead
7	O	sp^2-oxygen, carbonyl	34	Se	selenium
41	O	sp^2-oxygen, furan	35	Te	tellurium
47	O^-	carboxylate	59	Mg	magnesium
49	O	epoxy	61	Fe	iron (II)
69	O	amine oxide	62	Fe	iron (III)
70	O	ketonium oxygen	63	Ni	nickel (II)
5	H	hydrogen, except on N or O	64	Ni	nickel (III)
21	H	alcohol (OH)	65	Co	cobalt (II)
23	H	amine (NH)	66	Co	cobalt (III)
24	H	carboxyl (COOH)			

Figure 7: MM2(91) atom types developed by Allinger [J. Am. Chem. Soc., 99, 25, 8127-8134 (1977) doi:10.1021/ja00467a001] and reproduced from pg. 23 of [F. Jansen, Introduction to Computational Chemistry, 2nd edition, (2007) John Wiley & Sons Inc., Chichester, England]. Note that special atom types are defined for carbon atoms involved in small rings, such as cyclopropane and cyclobutane.

those of Class II) is necessary. However, if the interest is in studying the dynamics of a protein consisting of hundreds of amino acids (thousands of molecules), a crude model where whole amino-acid units are used as the fundamental type may be all that is achievable, considering the sheer size of the problem.

Figure 8 reports the main features of the most important force fields available to date and describes the most prominent functional forms used in these force fields.

The torsional energy is in all cases written as a Fourier series. Many of the force fields undergo developments, and the number of atom types increases as more and more systems become parameterized, and Figure 8 may thus be considered as a "snapshot" of the situation when the data were collected.

The Universal Force Field UFF —The combination of many atom types and the lack of a sufficient number of reference data have prompted the development of force fields with reduced parameters sets, such

Force field	Types	E_{str}	E_{bend}	E_{oop}	E_{vdw}	E_{el}	E_{cross}	Molecules
AMBER	41	P2	P2	imp.	12–6 12–10	charge	none	proteins, nucleic acids, carbohydrates
CFF91/93/95	48	P4	P4	P2	9–6	charge	ss,bb,st, sb,bt,btb	general
CHARMM	29	P2	P2	imp.	12–6	charge	none	proteins
COSMIC	25	P2	P2		Morse	charge	none	general
CVFF	53	P2 or Morse	P2	P2	12–6	charge	ss,bb,sb, btb	general
DREIDING	37	P2 or Morse	P2(cos)	P2(cos)	12–6 or Exp–6	charge	none	general
EAS	2	P2	P3	none	Exp–6	none	none	alkanes
ECEPP		fixed	fixed	fixed	12–6 and 12–10	charge	none	proteins
EFF	2	P4	P3	none	Exp–6	none	ss,bb,sb, st,btb	alkanes
ENCAD	35	P2	P2	imp.	12–6	charge	none	proteins, nucleic acids
ESFF	97	Morse	P2(cos)	P2	9–6	charge	none	all elements
GROMOS		P2	P2	P2(imp.)	12–6	charge	none	proteins, nucleic acids, carbohydrates
MM2	71	P3	P2+P6	P2	Exp–6	dipole	sb	general
MM3	153	P4	P6	P2	Exp–6	dipole or charge	sb,bb,st	general (all elements)
MM4		P6	P6	imp.	Exp–6	charge	ss,bb,sb, tt,st,tb, btb	general
MMFF	99	P4	P3	P2	14–7	charge	sb	general
MOMEC		P2	P2	P2	Exp–6	none	none	metal coordination
NEMO		fixed	fixed	none	Exp–6	quad, polar	none	special
OPLS	41	P2	P2	imp.	12–6	charge	none	proteins, nucleic acids, carbohydrates
PFF		P2	P2	imp.	12–6	polar	none	proteins
PROSA	41	P2	P2	imp.	12–6	polar	none	proteins
QMFF	32	P4	P4	P2	9–6	charge	ss,sb,st,bb, bt,btb	general
SDFF		P4	P4		9–6	charge, dipole, polar	ss,st,tt	hydrocarbons
TraPPE		fixed	P2	fixed	12–6	charge	none	C, N, O compounds
TRIPOS	31	P2	P2	P2	12–6	charge	none	general
UFF	126	P2 or Morse	$\cos(n\theta)$	imp.	12–6	charge	none	all elements
YETI	17	P2	P2	imp.	12–6 and 12–10	charge	none	proteins

Figure 8: Comparison of functional forms used in common force fields; the torsional energy reproduced from pg. 63 of [F. Jansen, Introduction to Computational Chemistry, 2nd edition, (2007) John Wiley & Sons Inc., Chichester, England]. $\mathrm{E}_{\text{torsion}}$, is in all cases given as a Fourier series in the torsional angle.

as the popular *Universal Force Field*, UFF developed by A. K. Rappe *et al.* [J. Am. Chem. Soc., 114, 25, 10024-10035 (1992), doi:10.1021/ja00051a040]. The basic idea in UFF is to derive $E_{\text{stretching}}$, E_{bending} and E_{torsion} terms from atomic constants (such as atom radii, ionization potentials, electronegativities, polarisabilities, etc.). The UFF and derivatives are claimed to be capable of describing molecules composed of elements from the whole periodic table, and these have been labeled as "all elements" force fields in Figure 8. For this reason, the UFF force field is expected to be less accurate compared with conventional force fields, but geometries of molecules and solids are often calculated qualitatively correctly. Relative energies, however, are much more difficult to obtain accurately, and conformational energies for organic molecules are generally quite poor.

Figure 9 gives an idea that force fields are constantly developed to improve their accuracy, their parametrization as well as their transferability among different systems.

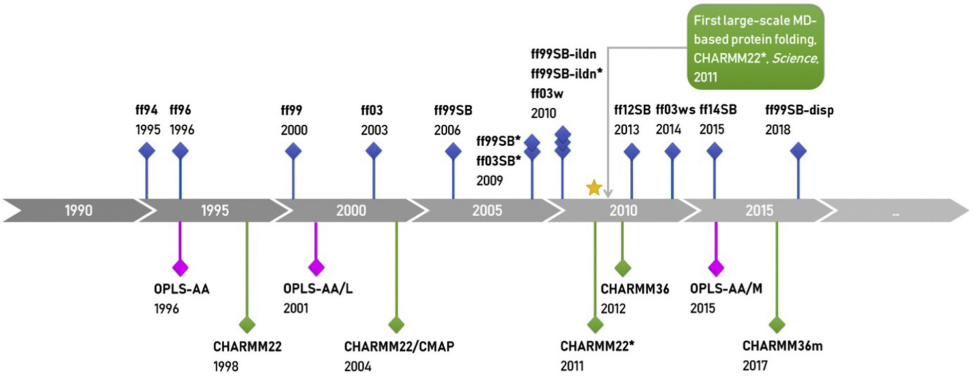

Figure 9: Time evolution of three most popular all-atom protein force fields, i.e., AMBER, CHARMM, and OPLS-AA [H. Geng *et al.*, *Comput. Struct. Biotechnol. J.*, 17, 1162-1170 (2019) doi: 10.1016/j.csbj.2019.07.010].

3.6 The Embedded Atom Model EAM

So far, we have discussed many ways to express parts of modern force fields (and interatomic potentials) used in the description of molecules and materials. However, force fields always require the underlying definition of bonds, and thus they may result unsuited to the description of metallic systems, whereby atoms interact with each other collectively in a many-body fashion.

An entirely different approach is that of the semi-empirical *embedded atom model* EAM, proposed by Murray Daw and Michael Baskes [*Phys. Rev. Lett.*, 50, 1285 (1983) doi: 10.1103/PhysRevLett.50.1285]. EAM is designed to treat purely metallic systems with no directional bonding, as in metals. The main physical property incorporated in the EAM is the moderation of bond strength by other bonds (coordination-dependent bond strength). Effectively, the EAM scheme sets its foundations on the fact that each atom can be embedded in a host lattice consisting of all other atoms. For example, in a metal, one can express the electron density of an atom at site i as an averaged atomic electron density of atom j via Eq. 3.10.

$$\rho_i^a = \sum_{i \neq j} f_j(x_i - x_j) \tag{3.10}$$

where f is the atomic electron density of atom j. Figure 10 shows graphically what is explained in Eq. 3.10. So, the key ingredient of EAM potentials is that the background density of each atom is determined by evaluating at its nucleus placed at position i the superposition of atomic-density tails from all other atoms js.

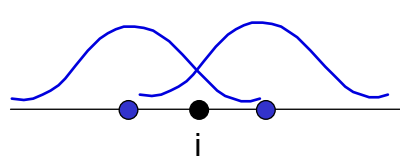

Figure 10: Overlapping of atomic densities of a reference atom i (black circle) by its neighbor atoms js (blue circles) as employed in the description of metals in EAM potentials.

Hence, in EAM potentials the total energy of a metal is assumed to be the energy obtained by embedding an atom into the local electron density provided by the remaining atoms of the system, and as defined in

Eq. 3.11.

$$E_{\mathrm{FF}} = \sum_i G_i \left[\sum_{j \neq i} \rho_j^a(x_{i,j}) \right] + \frac{1}{2} \sum_{i,j,j \neq i} V_{i,j}(x_{i,j}) \tag{3.11}$$

where G_i of Eq. 3.11 is the embedding energy term, ρ_j^a is the (empirical) spherically averaged atomic electron density of atom j at the location of atom i, and V is a pair-wise potential function of the Coulomb form (see Eq. 3.9).

The embedding energy G_i is defined as the interaction of the atom with the background electron gas. The functional G_i can be represented either analytically (see, for example, Eq. 3.12) or in Table form.

$$G[\rho] = G_0 \left[\frac{n}{n-m} \left(\frac{\rho}{\rho_e} \right)^m - \frac{m}{n-m} \left(\frac{\rho}{\rho_e} \right)^n \right] + G_1 \left(\frac{\rho}{\rho_e} \right) \tag{3.12}$$

However, the embedding function is typically tabulated to give an exact fit to the equation of state (Energy vs. volume). The atomic densities can be extracted from experimental data or fitting procedures of very accurate quantum mechanical calculations [e.g., E. Clementi and C. Roetti, *At. Data Nucl. Data Tables*, 14, 3-4, 177-478 (1974), doi: 10.1016/S0092-640X(74)80016-1].

Therefore, in EAM potentials each atom is embedded in a host electron gas created by its neighboring atoms. The atom-host interaction is described in a way that is inherently more complex than the simple pair-bond model. In this way, the embedding function incorporates some important many-atom interactions. It is possible to describe and understand interatomic interactions at defects in terms of either the embedding function or the effective many-atom interactions that arise from it. In particular, it is simple to demonstrate how bonding is affected by coordination. This naturally leads to an understanding of the difference between bulk and surface bonds, for example.

Within these constraints, the EAM provides a very useful and robust means of calculating approximate structure and energetics, from which many interesting properties of metals can be obtained. The EAM is currently the method of choice for performing semi-empirical calculations in close-packed metals; it combines the computational simplicity needed for larger systems (from thousand to million atoms) with a physical picture that includes many-atom effects and avoids some of the problems of the pair-potential models.

The EAM method has been applied successfully to bulk and interface problems, such as phonons, thermodynamic functions and melting points of metals, liquid metals, defects, grain boundary structures, alloys, segregation to grain boundaries, interdiffusion in alloys, and fracture and mechanical properties. The EAM has been applied to problems in surface structure, adsorbate phase diagrams, segregation to surfaces, surface structural order-disorder transitions, surface ordered alloys, surface phonons, and clusters on surfaces. For example, the table shown in Figure 11 and taken from [M. S. Daw, S. M. Foiles and M. Baskes, Mater. Sci. Rep., 9, 7-8, 251-310 (1993), doi: 10.1016/0920-2307(93)90001-U] shows the remarkable agreement of the predicted thermal expansion values α from EAM with the experimentally available data.

Element	α(EAM)	α(exp)
Cu	16.4	16.7
Ag	21.1	19.2
Au	12.9	14.1
Ni	14.1	12.7
Pd	10.9	11.5
Pt	7.8	8.95

Figure 11: Comparison of the linear coefficient of thermal expansion at room temperature computed using the EAM and experimental values (in units of 10^{-6} K^{-1}). Source [M. S. Daw, S. M. Foiles and M. Baskes, Mater. Sci. Rep., 9, 7-8, 251-310 (1993), doi: 10.1016/0920-2307(93)90001-U].

3.7 Useful Resources

Indeed, the execution of FF-based simulations requires simulation packages, many of which are still developed by research groups. However, a number of these FFs, codes, as well as visualisation tools, are available at these websites:

Resource for molecules and materials

- Comparison of force field implementations;
- NIST FF resource.

Resource for materials

- Large-scale Atomic/Molecular Massively Parallel Simulator;
- The General Utility Lattice Program: GULP This program is useful to fit many types of interatomic potentials from experimental data and data from high-level theory;
- Database of Published Interatomic Potential Parameters;
- Reactive Force Fields–ReaxFF by Adri van Duin;
- Embedded atom model potentials EAM for alloys.

Note, this list is far from being exhaustive.

4 Periodic Boundary Conditions in Simulations

Classical simulations, performed either with force fields (or interatomic potentials) or via first-principles quantum-mechanical methods (see notes of successive weeks) provide information about the properties of macroscopic systems. Yet, the number of degrees of freedom that can be handled is limited to only a few hundred to a few million atoms. Most simulations probe the structural and thermodynamical properties of a system of a few hundred to a few thousand particles, and such numbers are still far removed from the thermodynamic limit. For such small systems, it cannot be safely assumed that the choice of the boundary conditions (e.g., free or hard, or periodic) has a negligible effect on the simulated properties of the system.

Thus, to simulate the properties of bulk phases it is essential to choose appropriate boundary conditions that mimic the presence of an infinite bulk surrounding our n-particle model system, e.g. a large protein or a metal-organic framework or a surface of catalysts in contact with adsorbing gaseous molecules, etc. This is usually achieved by employing periodic boundary conditions PBC.

As shown in Figure 12, the volume enclosing the n-particles of a simulation is treated as the primitive cell of an infinite periodic lattice of identical cells.

A given particle (j, say) now interacts with all other particles in this infinite periodic system, that is, all other particles in the same periodic cell and all particles (including its periodic image) in all other cells. For example, if we assume that all intermolecular interactions are pairwise additive, then the total E_{FF}, of a system of n particles in any one periodic box is given by:

$$E_{\mathrm{FF}} = \frac{1}{2} \sum_{i,j,n}^{\prime} v_{i,j} \left(|r_{i,j} - nL| \right) \tag{4.1}$$

where L is the length of the cube enclosing the molecular system or materials under investigation, i.e. the periodic box. The prime sing over the sum indicates that the term with $i = j$ is to be excluded when $n = 0$. v can be assumed as any of the pairs between (atoms) represented by stretching-, bending-, torsional- and van der Waals-type interactions discussed in Section 3.

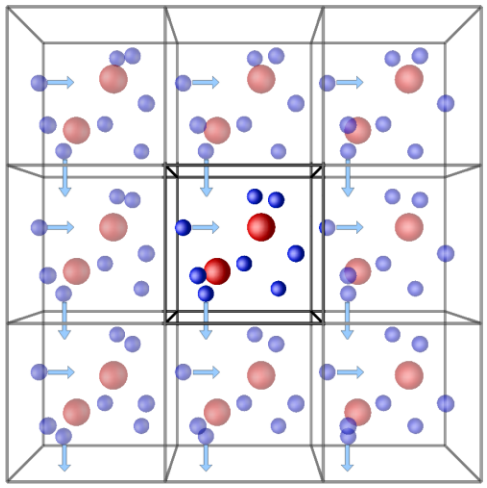

Figure 12: Schematic visualization of the idea of periodic boundary conditions, where an identical replica of the central cell are shown.

In this very general form, the PBCs are not particularly useful, because to simulate the behavior of the bulk material, we had to rewrite the potential energy as an infinite sum rather than a finite one. However, in practice, we are typically dealing with short-range interactions, and hence is possible to truncate all the intermolecular pair interaction beyond specific cutoff distances r_{cut}. Given a box size L, the maximum inter-atomic distance r_{cut} is equal to half of the edge of the simulation box, $\frac{1}{2}L$. When the minimum image criterion is satisfied, a particle can interact only with the closest image of any other particle as shown in Figure 13.

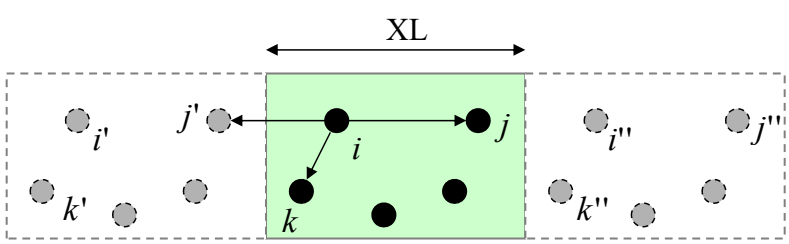

Figure 13: Visual representation of the PBC conditions along the x direction of a cubic cell. XL is L along the x axis. Particle indicated with $'$ and $''$ identified images across the boundaries. For example, the closest $i-j$ distance is not identified by the couple $i-j$ in the reference cell (green) but $i-j'$, which goes across the boundary.

Therefore, if a particle j is beyond the range of interaction with particle i (i.e., $r_{ij} > r_{cut}$), one has to find the closest images. A simple algorithm stemming directly from Eq. 4.1 to check for the closest image in a cubic cell is given in the pseudo-code below.

```
float dx = x_j - x_i;

if (x_axis){

   // half_x is half of the lenght_box
   float half_x = 0.5*lenght_box ;
```

```
  // check if the distance between atoms i and j
  // is greater than half the box size
   if (dx > half_x){
     dx = dx - lenght_box;
   }

  // check if the distance between atoms i and j
  // is lower or equal than the half the box size
  if (dx <= -half_x){
     dx = dx + lenght_box;
   }

}
```

Although the use of periodic boundary conditions proves to be a surprisingly effective method for simulating homogeneous bulk systems, one should always be wary that the implementation of PBC can cause spurious correlation effects not present in a truly macroscopic bulk system.

Criteria, rules, and algorithms to perform truncation of short-range non-bonded interactions, including van der Waals do exist but are not the objective of these classes.

5 Geometry Optimisation Schemes

So far we have discussed the strategies to compute the total energy of a particular structure of an arrangement of atoms in space, e.g., a molecule or a material. We found a suitable approximation of the total energy defining the total energy through a classical force field, hence $E_{\text{Total}} = E_{\text{FF}}$. However, E_{FF} is only meaningful if the position of atoms in the molecule or material under consideration is defined. In writing the expression of E_{FF} we have always imagined atoms to be "clamped" to their positions, e.g., we have always assumed x_0 in defining $\text{E}_{\text{stretching}}$ in Eq. 3.2. Therefore, a question remains: How can we find the optimal positions of atoms in space so that we can identify local and global minima in the PES (see Figure 3), and thus minimize E_{FF}?

A minimum (or a set of local minima) in the PES is identified by a set of atomic coordinates where the force $F(\vec{r})$ of Eq. 5.1, which is the force acting on atoms equal to zero.

$$F(\vec{r}) = -\frac{dE_{\text{Total}}}{d\vec{r}} \approx -\frac{dE_{\text{FF}}}{d\vec{r}} = 0 \tag{5.1}$$

where $\vec{r}$ is the coordinate system of reference x, y, z, and in Eq. 5.1 we have written $F(\vec{r})$ as the first derivative of the total energy E_{FF} with respect to $\vec{r}$.

So minima in the PES are those set of atomic coordinates (set of x_0, y_0, z_0) that minimize the E_{Total} (E_{FF}) concerning positions in space (reference system of coordinates x, y, z).
Importantly, for a molecule or a material with **N** atoms in its structure, the number of possible degrees of freedom that the optimizer algorithm will need to optimize in order to find a global minimum and local minima will be spanning a space $\mathbf{3N - 3}$.

In the following pages we will describe the three main methods utilized to explore the minima (i.e., minimize) in the PES.

5.1 The Steepest Descent Method

The minimization of $E(\{r^j\})$ in a space of variables r^j with $j = 1, N$ is a widely studied problem in mathematics, physics, and chemistry. In absence of any preliminary information about the PES of a molecule or material, the best choice for a direction of displacement from a point r_0^j (where 0 indicates a particular value of the variable r^j) to reach a minimum is the *steepest descent* (SD) direction:

$$\mathbf{g}_0^j = -\frac{\partial E_{\text{Total}}}{\partial r^j}|_{x^j = x_0^j} \tag{5.2}$$

where $\mathbf{g}$ is the gradient of E_{Total} with respect to a displacement of the atom (i.e., the force according to Eq. 5.1), and is shown by the initial direction point 1 in Figure 14.

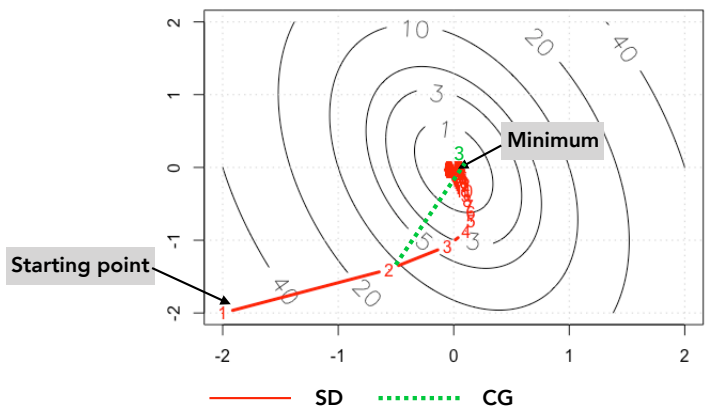

Figure 14: Schematic illustration of minimization schemes steepest descent (red solid line) and conjugated gradient (green dashed line) of E_{Total} in two dimensions $E_{\text{Total}}(\{x, y\})$. The steps $1, 2, 3 \cdots$, denote the SD or the CG steps. Numbers marking iso lines give a sense of the variation of E_{Total} far and close to the local minimum.

The lowest energy along the direction can be found by "line minimisation" in one-dimensional space, i.e., the minimum of E_{Total} as a function of the parameter α, where the new coordinate in the steepest descent r_2^j is defined as:

$$r_2^j = r_1^j + \alpha_1 \mathbf{g}_1^j \tag{5.3}$$

Of course, a series of such steps must be taken to approach the absolute minimum, generating a set of subsequent points $r_1^j, r_2^j, r_3^j, r_4^j, \cdots r_n^j$. This iterative process is illustrated in Figure 14 for a very simple function of two variables $E_{\text{Total}}(\{x, y\})$. We observe that even though the function E_{Total} decreases at each step, the step does not move directly to the minimum. One should note that the ability to minimize $E_{\text{Total}}(\{x, y\})$ in Figure 5.3 depends entirely on the choice of the parameter α_1. Hence, a bad choice of α (e.g., which causes a step to be too large) can drive the optimization away (escape) from the local minima. The SD method is particularly bad if the function E_{Total} has a very different dependence on the different variables so that the region around the minimum forms a long narrow valley, as that indicated by the label "minima" in Figure 3.

5.2 The Conjugated Gradient Method

Although it may seem surprising, there is a faster way to reach the minimum of $E_{\text{Total}}[\{x, y\}]$ of Figure 5.3 than to always follow the "downhill" steepest descent direction. After the first step x_i^0, one not only knows the gradient of $E_{\text{Total}}[\{x, y\}]$ (i.e., the first derivative, $g_i^0 = -\frac{\partial E_{\text{Total}}}{\partial r_i}|_{x_i = x_i^0}$) at the point x_i^0, but also have the value of the gradient at the previous point x_{i-1}. This additional information can be used to choose a

more practical direction along which the line minimization will lead to lower energy. In fact, for a quadratic function in $3N$ dimensions, the conjugated gradient (CG) method is *guarantee* to reach the minimum in $3N$ steps. We will consider this case explicitly to demonstrate the power of the CG method.

As the name of CG states the main ingredient in this optimization technique is to perform each line search (step) not along the current gradient (like in the SD method) but along a line that is constructed such that it is "orthogonal", here *conjugated*, to the previous search directions.

Then, the smartest way to make the subsequent step i is to move in the direction where the gradient along the previous direction $i-1$ remains zero, i.e. $\mathbf{g}_{i-1} = 0$. This condition can be achieved if the subsequent step is "orthogonal" to the previous one.

Therefore in CG, the first step is identical to as steepest descent step, but successive steps are performed along the line formed as a mixture of the current negative gradient and the previous search directions, enforcing the condition of orthogonality. If we define the derivative of the Total energy at step i of r_i as $\frac{dE_{\text{Total,i}}}{dr_i}$ the conjugated gradient method is written as in Eq. 5.4.

$$\frac{dE_{\text{Total},i}}{dr_i} = -\mathbf{g}_i + \beta_i \frac{dE_{\text{Total},i-1}}{dr_{i-1}} \tag{5.4}$$

where $\frac{dE_{\text{Total},i-1}}{dr_{i-1}}$ is the first derivative (gradient) of the Total energy of the previous step $i-1$, whereas β (at step i) is a value that needs to be set and updated at every step.

Several schemes have been proposed to select the value β are these are named after the inventors, e.g., *Fletcher-Reeves* (FR), *Polak-Ribiere* (PR) and *Hestenes-Stiefel* (HS).

$$\begin{aligned}
\beta_i^{\text{FR}} &= \frac{\mathbf{g}_i^{\text{T}}\mathbf{g}_i}{\mathbf{g}_{i-1}^{\text{T}}\mathbf{g}_{i-1}} \\
\beta_i^{\text{PR}} &= \frac{\mathbf{g}_i^{\text{T}}(\mathbf{g}_i - \mathbf{g}_{i-1})}{\mathbf{g}_{i-1}^{\text{T}}\mathbf{g}_{i-1}} \\
\beta_i^{\text{HS}} &= \frac{\mathbf{g}_i^{\text{T}}(\mathbf{g}_i - \mathbf{g}_{i-1})}{\mathbf{d}_{i-1}^{\text{T}}(\mathbf{g}_i - \mathbf{g}_{i-1})}
\end{aligned} \tag{5.5}$$

where $\mathbf{g}_i^{\text{T}}$ is the transpose of the gradient matrix including all the derivatives at step i of the $3N-3$ available degree of freedoms. $\mathbf{d}_{i-1} = -\frac{dE_{\text{Total},i-1}}{dr_{i-1}}$, and hence $\mathbf{d}_{i-1}^{\text{T}}$ is its transpose. Besides the complexity of β in Eq.s 5.5, the selection of these βs is important to solve pathological optimization tasks, whose discussion is beyond the scope of this class.

5.3 The Newton-Raphson Method

In contrast to the SD and CG methods, the Newton-Raphson (NR) technique expands the value of $E_{\text{Total}}(x_0)$ to the second order around the point (or a vector of points) in space x_0. This means that the second derivative of the energy versus the atomic is tracked.

$$E_{\text{Total}}(x) = E_{\text{Total}}(x_0) + \mathbf{g}^{\text{T}}(x - x_0) + \frac{1}{2}(x - x_0)^{\text{T}}\mathbf{H}(x - x_0) \tag{5.6}$$

where $\mathbf{g}^{\text{T}}$ is again the transpose of the gradient of $E_{\text{Total}}(x)$, whereas $\mathbf{H}(x - x_0)$ the negative of the second derivative of $E_{\text{Total}}(x)$, i.e. $-\frac{\partial^2 E_{\text{Total}}}{\partial {x_i}^2}$, also known as the *Hessian*. Thus the NR method requires the second-order term to become zero, which is achieved by setting the step search as:

$$(x - x_0) = -\mathbf{H}^{-1}\mathbf{g} \tag{5.7}$$

$\mathbf{H}$ is a square matrix containing the second derivatives of E_{Total} with respect to positions (thus the curvature of the PES), and if $\mathbf{H}$ is diagonalized one obtains the eigenvectors and eigenvalues ϵ_i. Using such eigenvectors and eigenvalues the NR step Δx_i can be written as in Eq. 5.8.

$$\Delta x_i = -\frac{f_i}{\epsilon_i} \tag{5.8}$$

where f_i is the projection of the gradient along the Hessian eigenvector with eigenvalue ϵ_i, which is the gradient component pointing in the direction of the ith eigenvector. Near a global minimum, all the Hessian eigenvalues are positive, and a step direction will be in the opposite direction to the gradient direction. However, if one of the Hessian matrix values is negative, the step in this direction will be along the gradient component, and thus increase of f_i to a positive value. In this case, the optimization will end at a stationary point with one negative eigenvalue of the Hessian, representing a first-order "saddle point" in the PES.

In general, the NR method will always attempt to converge to the nearest "stationary point", i.e. a local minimum. Thus, the NR method always performs well if one has an idea of what the minima should look like.